

LSTM based prediction of malaria abundances using big data

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ABSTRACT

Malaria prevails in subtropical countries where health monitoring facilities are minimal. Time series prediction models are required to forecast malaria and minimize the effect of this disease on the population. This study proposes a novel scalable framework to predict the instances of malaria in selected geographical locations. Satellite data and clinical data, along with a long short-term memory (LSTM) classifier, were used to predict malaria abundances in the state of Telangana, India. The proposed model provided a 12 months seasonal pattern for selected regions in the state. Each region had different responses based on environmental factors. Analysis indicated that both environmental and clinical variables play an important role in malaria transmission. In conclusion, the Apache Spark-based LSTM presents an effective strategy to identify locations of endemic malaria.

1. Introduction

Global warming is increasing the abundance and range of insect vectors of important human diseases, particularly in tropical countries [1]. Mosquito-borne diseases present a significant threat to health worldwide [2]. Ongoing efforts aim to control and ultimately eliminate endemic insect-borne diseases like malaria [3]. Many models have been proposed to predict malaria abundance and spread, from the early work of Christophers in 1911 [4] through to the present day. Each model has a different analysis strategy. Due to complex mosquito ecology and disease transmission cycle, the prediction of malaria is a challenging task [5]. Many researchers have developed prediction models based on environmental factors [6–8]. For malaria surveillance, predictions based on environmental factors and clinical conditions using big data have not been well-explored in the literature. India experiences a high incidence of malaria. In the early 1950s, India started the eradication of malaria. Since then, despite these efforts, 1000–1500 deaths have been reported yearly, and around one million cases have been registered [9,10]. The states of Telangana, Andhra Pradesh, Chattisgarh, and Orissa have the highest incidence of malaria in southeast India. We conducted our study in the state of Telangana in Khammam district (latitude 17° 14' N and longitude 80° 11' E). Khammam district has warm, with a humid climate and moist, dense forest which creates conditions favorable to malaria [11]. Malaria case abundance was studied at different geographic locations in Telangana, including Aswaraopeta, Khammam, Aswapuram, and Venkatapuram (Fig. 1).

2. Preliminaries

2.1. Big data

With the technological advancements in medicine come a huge volume of data generated from various sources in different formats, referred to as “big data” [12]. In the disease monitoring system, we store and process a very large volume of data. These data are a significant resource for analyzing diseases and making time-based decisions for public health [13]. Historical health records are analyzed to avoid curable deaths and predict the outbreak of diseases [14]. Disease pattern forecasting is useful for public administrations to investigate the outbreaks and instigate quick action to combat epidemics [15]. Big data in healthcare is widely investigated to improve timely disease diagnosis and facilitate best management of disease outbreaks. Big data has primary characteristics of volume, variety, velocity, value, veracity, and variability [16]:

- **Volume:** Refers to data generated in a large quantity from various sources such as satellite data, medical records, and lab reports, etc.
- **Variety:** Refers to structured, semi-structured, and unstructured data with various dimensions.
- **Velocity:** The data is generated with a speed which leads to new challenges in terms of management.
- **Value:** Represents the meaningful contribution of big data in making decisions.

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- **Veracity:** Refers to the reliability and quality of big data generated from various sources.
- **Variability:** This refers to the fact that data changes from time to time [17].

Technological advancement in healthcare systems has led to a massive amount of data, which is generated and stored in different locations. Predictive healthcare analysis can be made with the help of these systems [18]. Predictive analytics is used in the USA to improve understanding of diseases and optimize responses to disease [19]. Australia has made health-based apps for big data analysis to improve patient services and to detect insurance fraud [20]. Currently, studies to predict asthma use social media data, air pollution data, and sensor data with machine learning models. In general, research in predictive analytics using big healthcare data is generating considerable interest [21, 22]. The concept of big data is defined, in essence, as a large volume of data [16]. Big data contains both structured and semi-structured data. To handle such computationally complex data requires advanced, scalable tools such as Apache Spark and Hadoop MapReduce (Apache Software). But running computationally complex machine learning algorithms on Hadoop MapReduce is a time-consuming task [12]. Apache Spark is a hundred times faster than Hadoop MapReduce. However, unlike Hadoop, Spark does not come with its own file system and instead needs to integrate with another cloud-based file system [23].

2.1.1. Clinical data

The data were collected from all four geographical locations (Aswaraopeta, Khammam, Aswapuram, and Venkatapuram), from rural health centers (RHC) set up by the government of Telangana. These RHC were also used to gather the malaria samples from endemic places within Khammam district. A total of 52 RHC are monitoring malaria in Telangana, out of which 32 are in highly endemic areas. The staff of all RHC are well-trained in malaria diagnosis and testing. The primary data characteristics gathered by the RHC are the monthly average number of positive cases, cases without symptoms, average age, male, and female. The data used for this study were collected from January 1995 to

December 2018.

2.1.2. Environmental data

Environmental factors are such as rainfall, temperature, relative humidity, and vegetative index were also considered in predicting malaria abundance. The environmental data were collected from MODIS and TRMM satellites. The day and night time temperatures, from MODIS product MOD11A2 estimated an 8-day composite image at 1 km * 1 km spatial resolution. The daily rainfall, from TRMM product (TRMM3B42) were at a spatial resolution of 0.25° * 0.25° (27.8 km * 27.8 km). From the MODIS product (MOD13A1), 16-day composite processed images with an enhanced vegetative index were obtained at a resolution of 0.5 * 0.5. From the meteorological survey data acquisition center (MOSDAC) the daily relative humidity was recorded at a height of 2 m and 8 m above the ground.

3. Analysis

3.1. Artificial neural networks

Artificial neural network (ANN) systems are inspired by the structure of human and animal brains. An ANN allows parallel processing and interconnection, analogous to neurons in the nervous system. Mathematically, an ANN is a data-driven model that can solve complex nonlinear problems with a machine learning approach [24–26]. An ANN contains hidden nodes based on the number of input nodes specified by the programmer. The ANN model is trained by the stochastic gradient descent approach and updates the weights through backpropagation to minimize the errors. The loss function calculates these errors as the difference between observed results and actual results. In this process, the current error is backpropagated to update the weights and to calculate the loss function. It means this gradient tends to become smaller in backpropagation. This process leads to slow learning while training the ANN, which is a time-consuming process but also decreases inaccuracy [27].

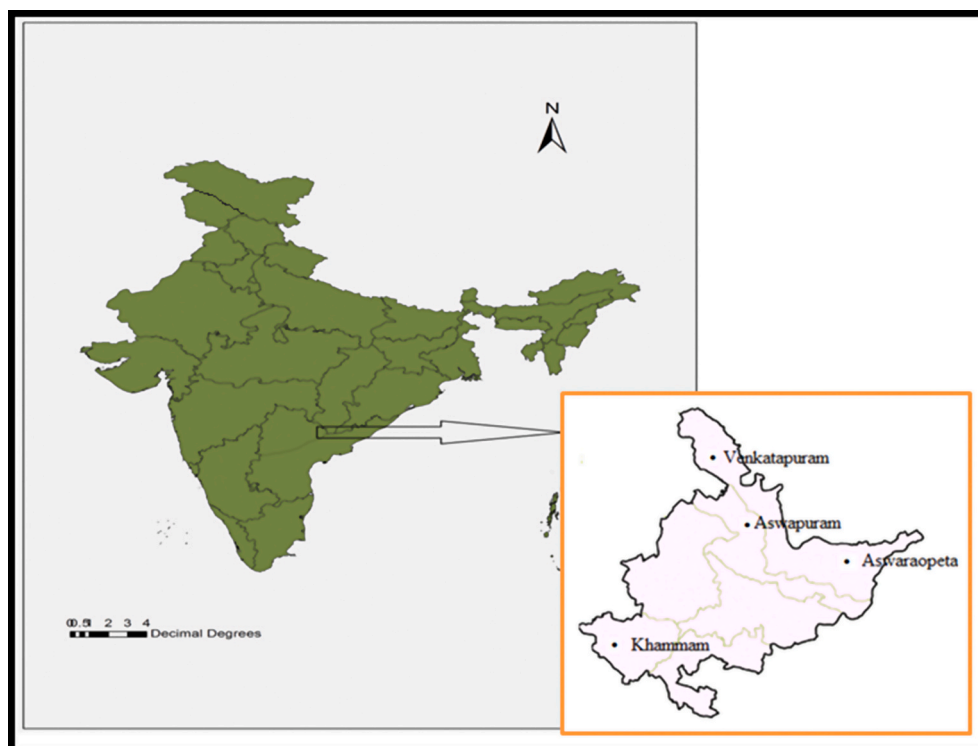


Fig. 1. Map of India showing different areas in the state of Telangana selected for study [31].

3.2. Recurrent neural networks

Recurrent neural networks (RNN) consist of a chain-like structure with one input node and one or more hidden nodes. The idea behind the RNN is memory. In this model, the memory will store the information from the previous steps. The RNN contains a feedback loop to update the information from the sequence of inputs in the learning process. While training the RNN, backpropagation is applied to calculate the gradients and weight adjustments. However, RNN updates the weights with a feedback modification process. The information is continuously repeated to update the weights makes the model unstable. Therefore RNN is not efficient for training in the long term [28].

3.3. Long short-term memory

To overcome the problems of speed and stability in RNN, long short-term memory (LSTM) was proposed by Hochreiter [29]. LSTM is very similar to RNN, but in this model, a new concept is introduced with one cell or interaction per module. As presented in Fig. 2, LSTM is a chain-like structure capable of remembering information and long term training with four network layers. The LSTM network consists of memory blocks like cells. This cell state is an important state which allows the data flow forward and remains unchanged. However, the data can be added or removed with sigmoidal activation functional gates. These gates are a series of matrix operations with unlike specific weights. With the implementation of gates, long-term dependency problems can be avoided while memorizing the LSTM.

The step by step process of LSTM is as follows:

The first step of LSTM is to identify the data that is not required for the process, and it is performed by the sigmoid function and takes the

output as (v_{t-1}) at time $t-1$ and current input (X_t) at time t . From the old output, the sigmoidal function decides which part of the output must be removed. This is called the forget gate (f_t). Here f_t are the vector values lie between 0 and 1, matching to each number in the cell, C_{t-1} .

$$f_t = \sigma(w_f[v_{t-1}, X_t] + b_f) \quad (1)$$

Here σ is the sigmoidal function, w_f is the weights and b_f is the bias of the forget gate.

The following equations have two states: one ignoring state, and one storing state from the current input X_t in cell state. The process contains two layers: the sigmoidal layer and the tanh layer. The sigmoidal layer decides if new information is updated or ignored using 0 or 1. In the second layer, the tanh update weights and passes the values between (-1 to 1). Values are selected according to their level of importance. As presented in Eq. (4), both the values are updated and form the new cell state.

$$m_t = \sigma(w_m[v_{t-1}, X_t] + b_m) \quad (2)$$

$$N_t = \tanh(w_n[v_{t-1}, X_t] + b_n) \quad (3)$$

$$C_t = C_{t-1}f_t + N_tm_t \quad (4)$$

Now in the final step, the output v_t is based on the output of sigmoidal gates (Q_t) and is multiplied with the new layer created by $\tanh$ C_t .

$$Q_t = \sigma(w_q[v_{t-1}, X_t] + b_q) \quad (5)$$

$$v_t = Q_t \cdot \tanh(C_t) \quad (6)$$

Here, w_q and b_q are the weights and bias, respectively, for the output

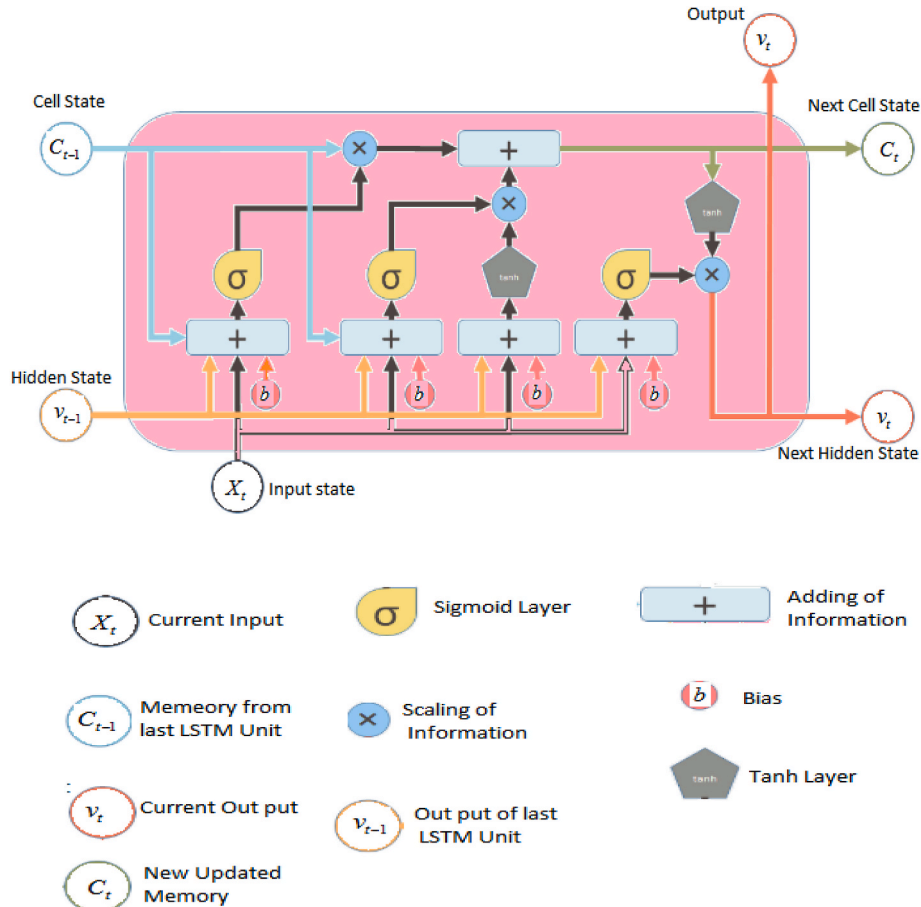


Fig. 2. Structure of a long short-term memory(LSTM) neural network. Reproduced from Ref. [30].

gates.

3.4. Model

Environmental variables such as rainfall, relative humidity, temperature, and vegetative index and clinical data with the number of asymptomatic cases were considered as input to predict malaria. We collected average daily rainfall, average daytime and nighttime temperature, and average enhanced vegetative index at 1 km resolution. The data were then aggregated as a monthly frequency. The dataset contains around 11% missing variables, and this problem can be handled with multivariate imputation by chained equations in data pre-processing. The clinical and environmental variable data were taken from January 1995 to December 2018 at each location (Aswaraopeta, Khammam, Aswapuram, and Venkatapuram) shown in Fig. 1.

The pre-processed data from January 1995 to December 2018 were used to train the model. The model was adjusted after certain iterations.

$$\text{Accuracy} = \frac{(TP + TN)}{(TP + TN + FP + FN)}, \text{Specificity} = \frac{(TN)}{(TN + FP)}, \text{Sensitivity} = \frac{(TP)}{(TP + FN)}$$

$$\text{Precision} = \frac{(TP)}{(TP + FP)}, \text{F-Measure} = 2 * \frac{\text{precision} * \text{Recall}}{(\text{precision} + \text{Recall})}$$

Finally, the model was tested for the period January 2019 to December 2019 and generated a monthly forecast of malaria abundances for 12 months, based on historical data of environmental and clinical variables.

Forecasts were generated for each geographical location and corresponding model performance measures were calculated. The malaria prediction framework was implemented in the Apache Spark framework (Fig. 3). Apache Kafka was selected as a message broker to place the data in Spark for batch processing and in real-time streaming.

The experiment was divided into two phases: i) offline batch processing ii) online streaming. In the first phase, the data was pre-processed and used for training. In the second phase, we performed the real-time analysis collected from the message brokers in batches. In detail, initially, the refined environmental and clinical time series data of each location was pre-processed by applying multivariate imputation by the chained equation to handle the missing data points. Then training was performed on the LSTM classifier. In the second phase, testing was performed for real-time analysis, and the results were forecast for a certain period. Final results were stored on the Hadoop distributed file system for online monitoring.

4. Results

The studied environmental and clinical variables represent a time series and are seasonally dependent. For example, the vegetative index and number of asymptomatic cases are time series [32]. The number of asymptomatic patients and their treatment were included. In the current study, predicted malaria instances for 12 months from January 2019 to December 31, 2019 are shown in Fig. 4. Of the four locations, Venkatapuram and Aswaraopeta recorded the highest number of cases (Fig. 4A–B). At these locations, the average rainfall was 1.6 mm, and the average daytime and night-time temperatures varied slightly by 3 °C compared with the other locations. The vegetative index also varied slightly at these locations. Khammam and Aswapuram have a lower number of recorded cases (Fig. 4C and D).

The malaria prediction classifier performance is evaluated using five different metrics:

Where True Positive = TP, True Negative = TN, False Positive = FP, False Negative = FN.

5. Discussion

Precise public health prediction models are useful for decision making and planning resources for better and more efficient services. In this study, we observed that environmental and clinical variables played a significant role in the prediction of malaria disease. This is the first attempt at combining clinical and environmental big data to predict malaria using LSTM on the Apache Spark platform, to best of our knowledge.

To update the different values in the internal cell states, we considered different training options and parameters and dynamically adjusted to get the best results. The adaptable parameter updated in the neural network, depending on the given loss function of the LSTM iteration. To allow learning of the dependencies effectively in the LSTM model, we changed the format of the input data layer. Instead of placing the data as an input data vector, we arranged the data in sequential lengths with their time stamps. This may provide information about the changes in trends, forecasted values, and their timestamps.

However, the sequential length of the data depends on the data type and is chosen by the programmer. The effect of subsequent data on the predicted values depends on the observed frequency in a time series problem [33]. For example, in environmental data, the values of the variables collected today may change on a different day. Theoretically, when the size of the sequential length increases, the LSTM model can better capture changes in the trend. In LSTM, an increase in sequential length may consume more time and resources while training the complex neural model. This may lead to constraints in ANN and RNN. The value of sequential length may vary from 5 to 10 in our study. In these cases, we observed no changes in the execution and the model performed well.

Another challenge in this study was adjusting the training parameters such as learning rate and number of epochs while training the LSTM. The best-optimized parameter was selected from Adam [34]. This method is widely used in deep learning applications and natural language processing. Instead of using Adam's default learning rate value

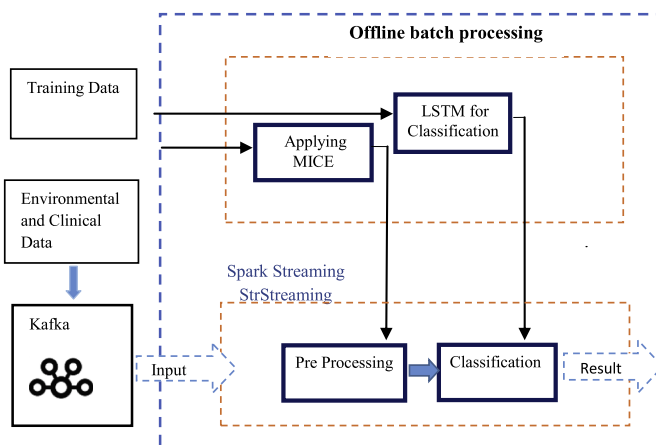
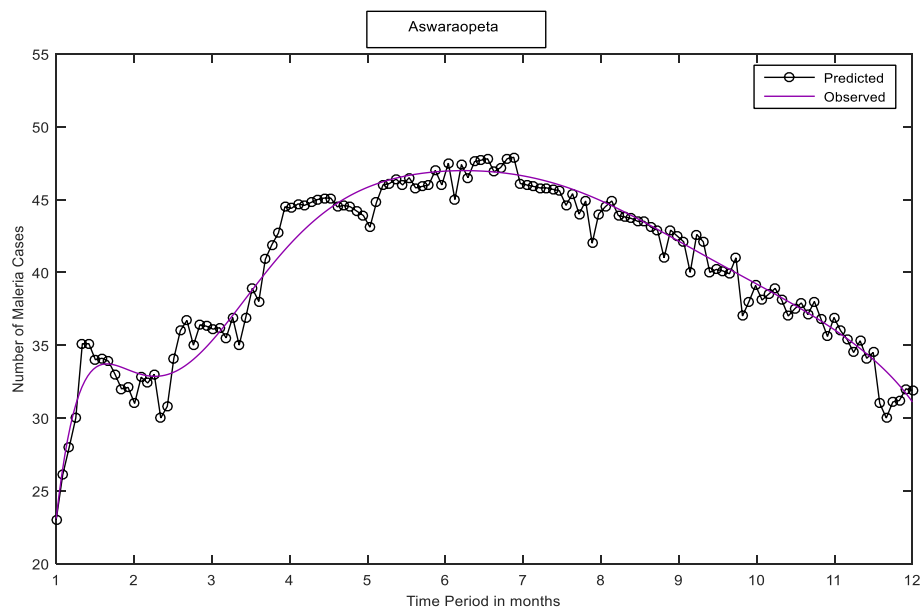
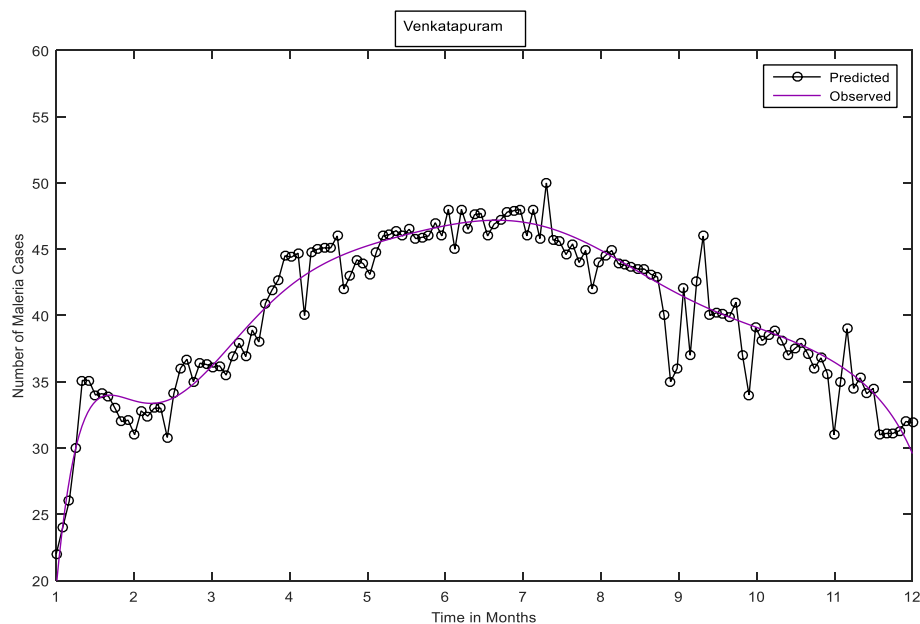


Fig. 3. Spark-based LSTM framework.



(A). Predicted and observed malaria instances for *Aswaraopeta* Area



(B). Predicted and observed malaria instances for *Venkatapuram* Area

Fig. 4. Predicted malaria instances for all studied geographical areas. Predicted and observed malaria instances for (A) Aswaraopeta area, (B) Venkatapuram area, (C) Khammam area, and (D) Aswapuram area.

(0.001), we have used a smaller value (0.00001). A small value in the learning process makes the training process slower. The selection of the number of epochs was also optimized. Generally, one epoch is considered as a single pass of an entire dataset forward and backward through the neural network. During the training process, a maximum of 100,000 epochs is recommended. They are many methods to improve model efficiency. One of the most efficient methods is to stop training early to prevent overfitting [35]. The early stopping method is widely applied in LSTM models. The LSTM model can be stopped after 2000 epochs if the model does not improve the results.

Across the studied geographical areas, there were individuals with malaria with no symptoms, and this result was confirmed with lab tests

that have high predictive power. Apart from the clinical variables, environmental variables such as rainfall, temperature, relative humidity, and vegetative index also play a significant role in prediction. The results of the prediction model varied from location to location, as presented in Fig. 5. Venkatapuram recorded the highest number of cases, and Khammam has the lowest number of cases. Fig. 6 represents the location of Khammam district, and the color of the map indicates the study locations. The accuracy of the model also varies from location to location. The classification results for the testing data at different locations are presented in Tables 1 and 2. Venkatapuram had the highest classification accuracy compared with the other locations. Khammam had the lowest accuracy as well as the lowest number of malaria cases.

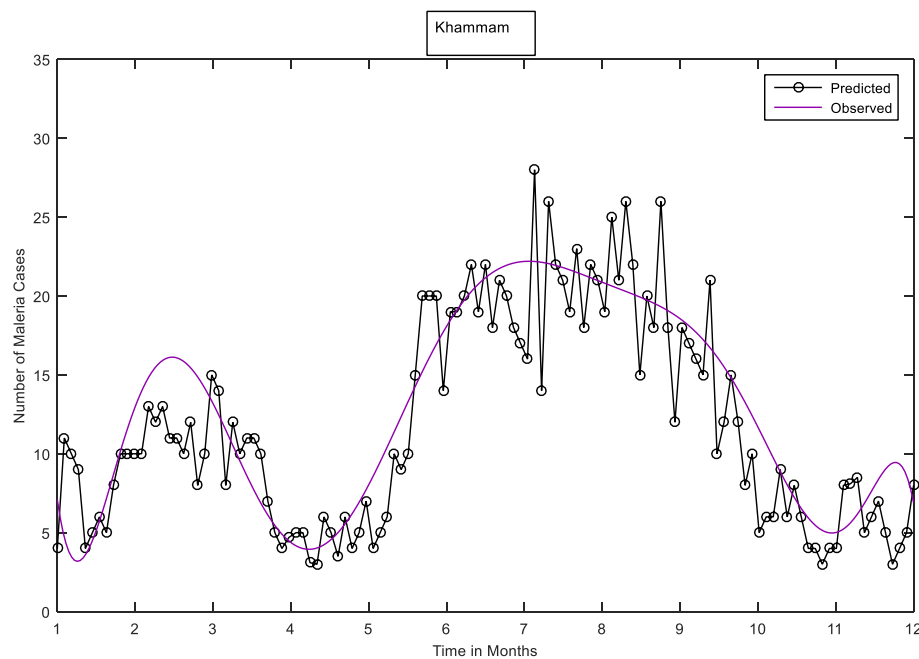
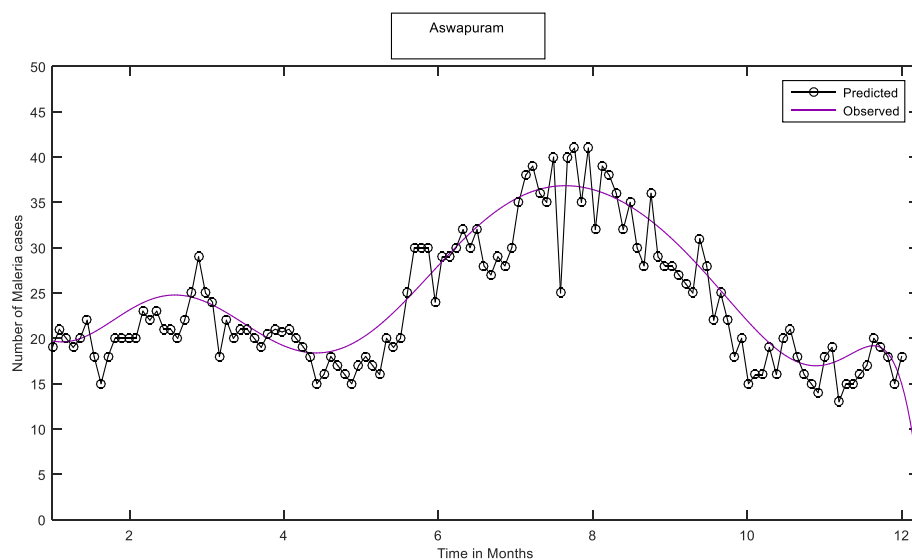
(C). Predicted and observed malaria instances for *Khammam* Area(D). Predicted and observed malaria instances for *Aswapuram* Area

Fig. 4. (continued).

One of the most interesting observations in this study is that most of the malaria cases were recorded between August to November. From the environmental data, rains generally start in the Telangana region in June and July. Hence, malaria cases increase after July and peak in August and September. Humidity and temperature play a significant role in malaria spread. When the temperature is 28 °C and humidity is 60% these conditions are favorable for breeding of the *Anopheles* mosquito vectors [31]. Conversely, when the temperatures are above 30 °C, and below 16 °C there is a negative impact on *Anopheles* mosquito breeding. We also observed that the enhanced vegetative index plays a significant role in the transmission of malaria. Venkatapuram, Aswawapuram, and Aswaraopeta have the highest number of cases with a high vegetative index. Khammam has a low number of cases and a poor prediction because of a low vegetative index.

Disease prediction models are useful in planning the resources where

medical facilities are limited in practice. The medical staff involved in malaria treatment could plan targeted visits to endemic areas. With the help of these models, they could provide appropriate treatment and diagnoses to minimize the impact of the disease in humans. In the future, we aim to exclude environmental data and consider only medical big data records for the prediction of malaria case numbers.

6. Conclusion and future scope

In this study, we have presented malaria disease patterns in different locations in Khammam district, Telangana State, India. We observed that environmental variables play an essential role in enabling the presence and transmission of malaria. Previously, several machine learning models have been developed to address malaria, but to our knowledge, this is the first attempt to forecast malaria case abundance

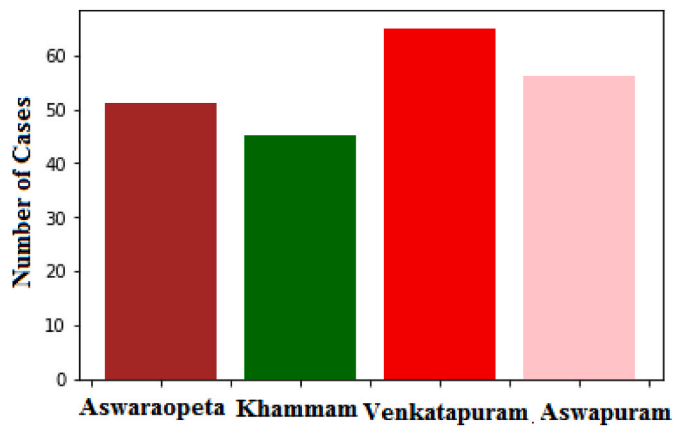


Fig. 5. Predicted malaria cases from January 2019 to December 31, 2019.

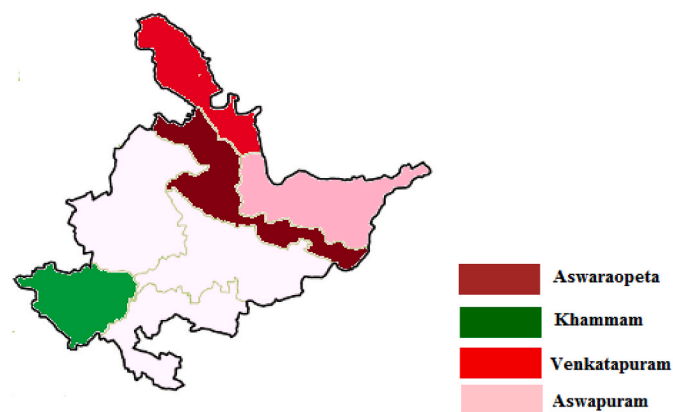


Fig. 6. Khammam area in the State of Telangana chosen for the study.

Table 1
Numbers of malaria cases in different locations.

Location	Minimum	maximum	mean	Standard deviation	mode
Venkatapuram	22	50	39.87	5.868	46
Aswapuram	13	41	23.61	3.214	20
Kammam	3	28	11.81	6.603	10
Aswaraopeta	23	47.9	40.3	5.667	46

Table 2
The performance measures of the proposed model for malaria case number prediction.

Location	Accuracy	Specificity	Sensitivity	Precision	F-Measure
Venkatapuram	96.11%	97.42%	95.20%	95.44%	96.70%
Aswapuram	94.42%	95.22%	92.89%	93.11%	94.49%
Kammam	88.72%	89.31%	87.23%	87.69%	88.91%
Aswaraopeta	93.26%	94.13%	92.94%	92.87%	93.68%

with environmental and clinical variables using LSTM on big data platforms. The proposed model is capable of handling and analyzing the massive amount of data with low latency and scalability. In future work, to understand malaria in different dimensions, we would analyze only clinical big data incorporating drug-resistant malaria strains. Precise predictive models are essential for early warning and for allocation of medical resources to reduce the effect of the disease outbreaks. Further investigation is necessary for malaria prediction to bring machine learning models into real-time practice.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compbiomed.2020.103859>.

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